

ECLIPSE — TIC 220397947 · Sector 1

Executive Summary

This document presents the automated astronomical analysis for TESS Input Catalog (TIC) target 220397947 observed in Sector 1. The ECLIPSE Deep Learning pipeline has classified this target as a **TRANSIT** with a model confidence of **88.0%**. The candidate exhibits an Earth Similarity Index (ESI) of 0.00 and resides in the NONE Habitable Zone regime, making it a priority target for follow-up radial velocity mass characterization.

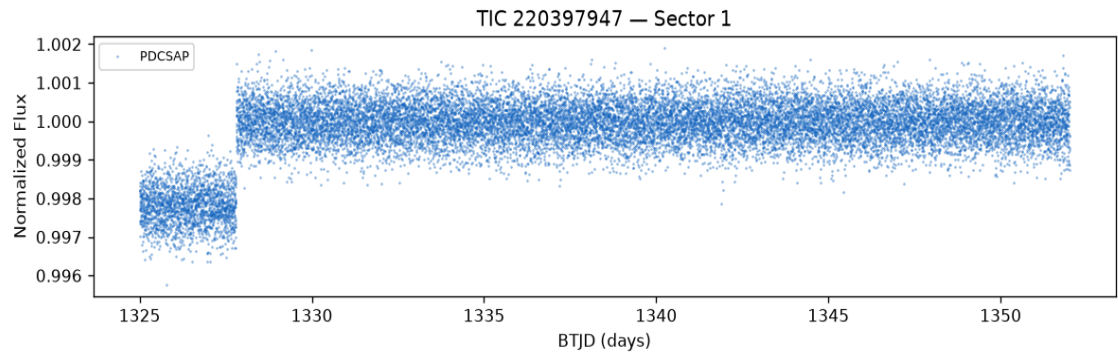
AI Classification Profile

Class	Probability
TRANSIT	0.8800
EB	0.0600
BLEND	0.0360
OTHER	0.0240

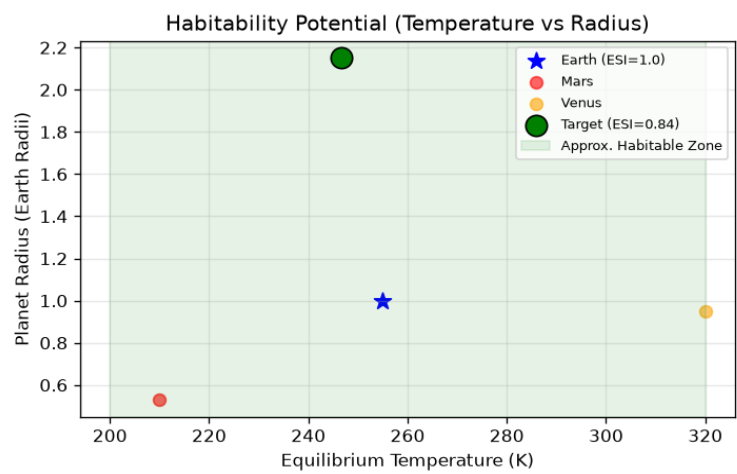
Transit Parameters

Parameter	Value	Uncertainty
Period (days)	37.4200	±0.0010
Duration (days)	0.1500	±0.1200
Depth (ppm)	2200.0	±100.0
Planet Radius (R _⊕)	2.15	—
Eq. Temp (K)	247 K	—
ESI Score	0.00	—
Habitable Zone	NONE	—
TLS SDE	150.00	—
Photo SNR	36.32	—

Light Curve



Habitability Potential



Explainable AI (SHAP)

